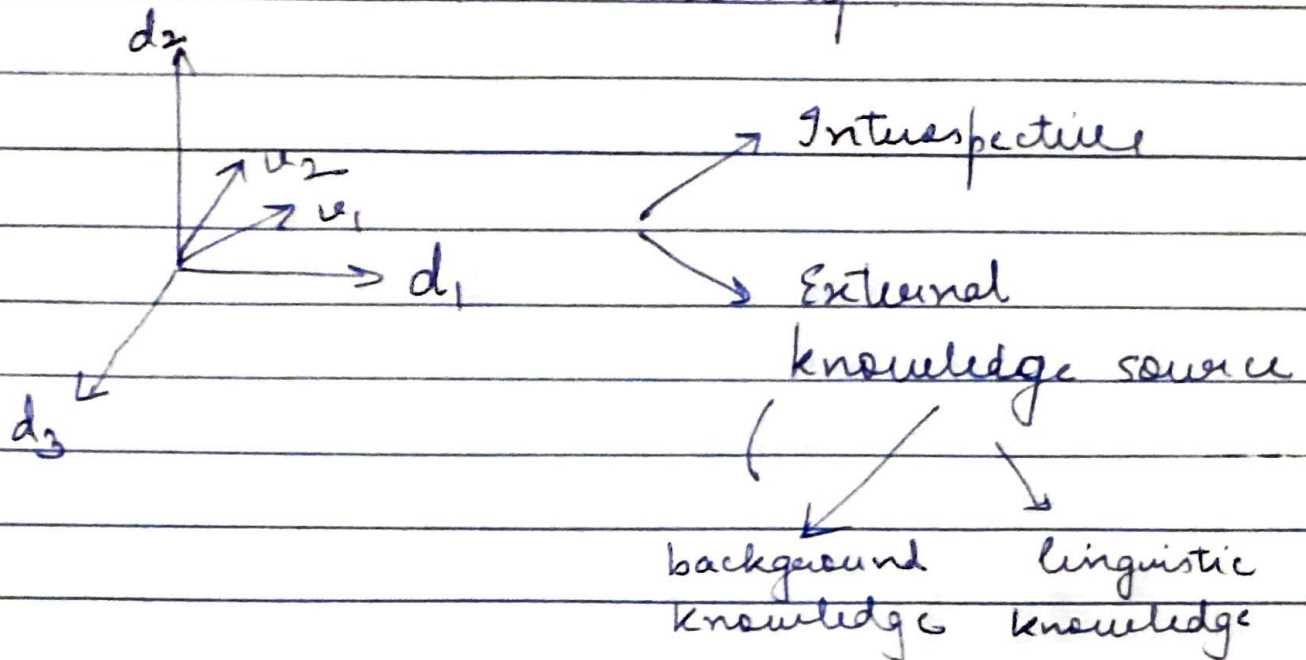


26th Feb 2026 / CS6370

→ (contd.)

- how to break the circularity?



introduces the idea of "concepts"

↓
then the semantic relationships are established by "concepts" & not "words"

↓
build a concept map

dictionary (representaⁿ of words at certain level)

↓
Psycholinguistic lexicon (WordNet)

↓
how to arrive at a number for the measure of relatedness

- word refers to sth. in reality. Straightforward mapping of words to objects.

- theory of structuralism → straightforward idea is incorrect. Poverty of stimulus argument (we don't know enough of abt. the lang. to build the concepts).
- humans are by design meant to understand the lang. Any amt. of data cannot fix that fundamental problem.
- concepts can be defined only in terms of one another.

meaning \ Features

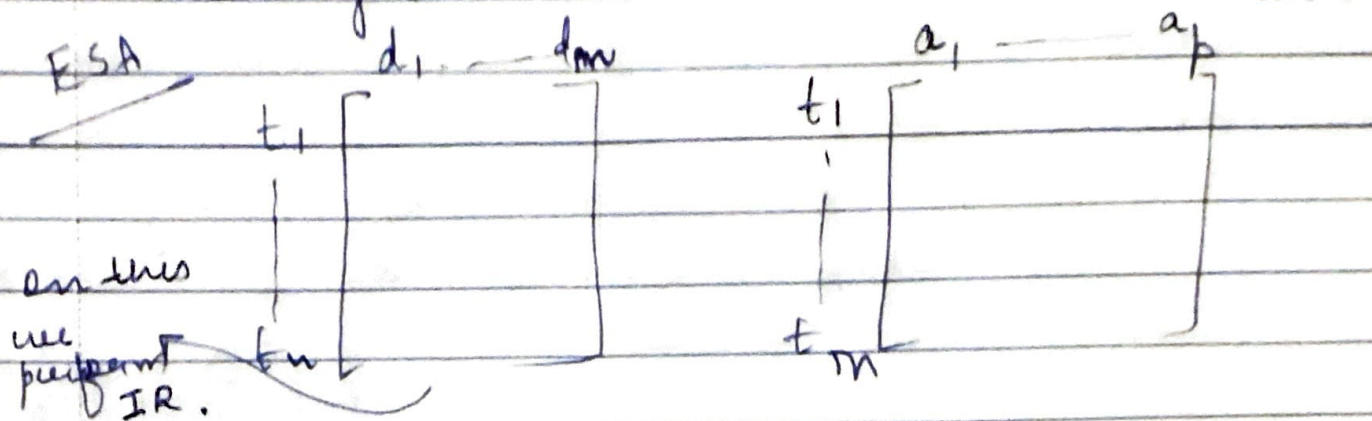
Synonym set (synset)	S1			
	S2	✓		✓
	S3	✓		

polysemous

from here we can build the concept network.

- synsets have semantic relationships.
- Antonyms are mainly based on the word forms and not on the synsets.
- lexical relationship.
- linguistic knowledge
- ESA (Explicit Semantic Analysis)
- relation btw. words & related words

- the background knowledge is input. It uses Wikipedia for ESA algo. A models used meanings.



- now I'm breaking circularity, since my "abstract" comes from $t \times a$. Every term is a linear combination of these concepts.
- Articles are orthogonal to each other.
- All these structural & semantic relationships, only work at concept level. We lose these relationships, once we move to the no. phase.